AHM Statement on OpenAI's October 6 Release of Mathematical Documents

7 October, 2026 in guest blog, opinion | Tags: AI, Association for Human Mathematics | by Terence Tao

[This is a guest post by the Association for Human Mathematics, reposted from their statements page. This blog post was initially written in a different file format and converted using AI. — T.]

Yesterday, on October 6th, 2026, OpenAI — which is currently defending lawsuits against accusations of illegal plagiarism, copyright infringement, and trademark dilution — released a repository of manuscripts purporting to contain solutions to a number of high-profile problems in mathematics.

Mathematicians did not ask for this work to be done. The Advisory Group on Mathematics and Artificial Intelligence, from whom OpenAI has claimed to derive its legitimacy, opened their initial advisory statement by saying that frontier AI corporations should not test advanced mathematical problems on internal models. In ignoring the central premise of the Advisory Group's position, OpenAI has indicated total disregard for the norms of scientific research — norms that guarantee that mathematics remains trustworthy, ethically researched, and in the public interest.

Mathematicians have a particular vision of progress that is informed by history and field-specific considerations. We reject OpenAI's assertion that this release advances our subject, and we urge mathematicians and the public to view the value of this publication model with due skepticism.

Releasing over 700 files at once is not a demonstration of scholarship, but a demonstration of power. We urge mathematicians to discontinue their work with OpenAI and to return to a vision of science that centers human understanding.

Association for Human Mathematics Communications Working Group

152 comments

Comments feed for this article

8 October, 2026 at 6:34 am **ryeguy10**

I think it would be good if you edited this to put a comment at the top saying that you are not a member of the AHM and this is not your personal opinion, but one among many perspectives in the math community, and that you are committed to pluralism and publishing a variety of perspectives.

A false narrative is now developing in many circles about how mathematicians are extremely anti-AI, and there is lots of misinformation on X circulating about you are supposedly aligned with the AHM despite not being a member and posting previous blog posts that are pro-AI, like the ones by Amit Sahai and Dimitris Koukoulopoulos.

I think this false narrative is extremely harmful to the future of mathematics. The fastest way to get a Republican administration and leaders in industry to want to defund mathematics is to convince them we all

extreme anti-AI luddites. The current administration is unfortunately pretty hostile to science funding. This will exacerbate it.

👍 169 👎 56 Rate This
Reply

8 October, 2026 at 7:05 am
Anonymous



Don't worry: Musk, Altman, Sacks, Kushner etc. know exactly who is pro or anti AI.

There is a general problem though on other discussion forums that people do not understand the meaning of "guest post" and attribute everything to Tao (even though "guest post" features prominently already).

👍 27 👎 8 Rate This
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8 October, 2026 at 11:28 am
ryeguy10



The misinformation appears to be corrected:
https://x.com/littmath/status/2108182724469014833

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8 October, 2026 at 10:52 am
Anonymous



I think the fastest way possible to defund mathematicians is by convincing the Republican administration and leading figures in the industry that mathematicians are useless idiots. Personally, I would prefer to be seen by these people as the peaceful reincarnation of Ted Kaczynski rather than some useless idiot.

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10 October, 2026 at 7:22 pm
ryeguy10



Absolutely nobody is going to be convinced that mathematicians are useless idiots. Even if mathematics becomes like chess, which I don't think will happen anytime soon, this will still not happen. Nobody thinks chess players are useless idiots. In fact, more people play chess now than before Deep Blue.

👍 1 👎 1 Rate This
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9 October, 2026 at 4:26 am
pascoej



You sound like a gangster saying you will turn over Terry Tao's shop when you talk like that.

👍 4 👎 11 Rate This
Reply

9 October, 2026 at 9:25 am
ryeguy10



I am very much both unwilling and unable to do such a thing.

👍 9 👎 0 Rate This
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10 October, 2026 at 6:54 am
pascoej

I understand, but the notion that if you don't comply politically the whole field will die is dangerous nonsense- we need to be reasonable and respect each others choices about AI, just in case the other people are right, etc.

0 0 Rate This
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10 October, 2026 at 7:20 pm
ryeguy10

To be clear I certainly respect others' choices about AI. Personally I am currently starting a collaboration with a mathematician who has much more negative feelings towards AI than I do. We will not use AI at least at the beginning. Obviously it's very important to value pluralism and free speech. Some people feel positively about AI (like me) and others don't. Some people really enjoy using AI and some don't. That's good! Of course it's good to have different kinds of people doing different kinds of things. Each approach has its advantages and disadvantages and we would benefit from diversification.

However I think freaking out about AI is both (1) wrong, and (2) damaging to the profession. I talk about (2) because I don't think I can convince people of (1) who don't agree with it already, and because a lot of the freaking out is driven by fear that AI will damage the profession. But I think what will really damage the profession is if mathematicians are against mathematical progress.

I completely agree that human achievement and human understanding are also valuable and we should communicate that we value those things too, but we can do that in a way that is not being sad that a theorem was proved. If you make a conjecture it's good if it's proved. If it is proved in a silly Goodharted way that doesn't teach you anything about the world, you can go make other conjectures instead that require new and interesting ideas to solve, and you learn what the "right" conjecture is!

0 0 Rate This

9 October, 2026 at 2:37 pm
Anonymous

For many among an entire generation of budding mathematicians with nascent research programs at the cutting edge of the field, the appropriate direction for their work has now been rendered significantly unclear, and choosing new directions for research is now hyper-complicated by the borderline impossibility of the difficulty-for-current-AI / difficulty-for-AI-in-two-weeks / difficulty-for-me ratio calculus. This could have been significantly avoided or delayed by OpenAI (and any other moral actor with access to such models) not publicly sharing these results, and simply holding onto them. For Pete's sake, has anyone ever heard of the Prime Directive? Presumably OpenAI would argue that someone would have developed such technology and acquired many such proofs in one or ten years anyway. The problem with that argument is the premise according to which inevitability justifies immediate release. The Prime Directive is fiction but instructive: had people had more time to weigh the costs and risks of a career in mathematics in light of the development of such technology before OpenAI's large dumps of proofs, mathematical society—comprising all those individuals weighing their career options—could acclimate and avoid the harms of the sudden avoidable jolt. Likewise, when you buy a pet fish, you don't just dump it in the enclosure, you let the bag sit for a while.

2 4 Rate This
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10 October, 2026 at 3:28 am
ryeguy10

I prefer immediate release. I think it's way worse for the labs to sit on solutions to problems. It's not fun to not know if the problem you are working on is already solved. I do agree though that nowadays we should put a much higher value on coming up with a second proof of an existing result.

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10 October, 2026 at 7:08 am
pascoej

The thing is, we discovered math warp drive, so we are not protected by the Prime Directive. We have to respect those finding their starship crew and those with the desire to be planet people.

👍 0 👎 0 Rate This
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10 October, 2026 at 7:28 pm
ryeguy10

What exactly are the costs and risks of such a career? How exactly would academic mathematics end? What is the most plausible scenario where this happens? The only scenario I can envision is that if the people who fund mathematics – governments, former mathematicians in finance and in hi-tech – stop funding us. Thus I think the main risk to mathematics as a profession is things that will cause these outside actors to defund us. Some of the stuff said by Buckmaster, Hugo Duminil-Copin, and unfortunately Terry himself (Terry if you are reading this, I obviously know this is not your intent) is thus very threatening to the future of the profession. I don't think Buckmaster or Terry are anti-AI luddites at all but I think they should be a bit more careful. If all we talk about is we are upset now because AI solved our grant problems, no one will want to fund us.

Mathematicians are very curious people and we will find stuff to do. Come up with the shortest and most elegant proof of an existing result. Come up with a new conjecture and describe why it is outside the range of existing techniques. And so on and so forth. If we talk about all the exciting new possibilities this opens up, which we should, then people will definitely want to fund us. I think humans+AI>AI alone for quite some time. And I think academic mathematicians can certainly persist even if math becomes like chess.

👍 0 👎 0 Rate This
Reply

8 October, 2026 at 6:35 am
Jörg Neunhäuserer



I worked on three of the problems that have now been solved by ChatGPT (153, 148, and 146), and I spent quite a lot of my life trying to find proofs for them. Am I now frustrated or disappointed, feeling that AI has taken something away from me? No, not at all, actually. I'm rather relieved that I no longer have to do this work, and I have to admit that I'm also a little proud that my mathematical intuition turned out to be right in all three cases.

But there is another aspect to this. The texts produced by AI do not represent human mathematical understanding. Human mathematical writing, with all its quirks and individuality, is something quite different: it is an expression of a person's own understanding.

Should I now turn the AI's texts into something more like that? Perhaps. Or I could describe new abstract objects and formulate conjectures about them, which an AI could then prove. Probably the best approach is to do both!

👍 157 👎 19 Rate This
Reply

8 October, 2026 at 7:58 am
Anonymous



"problems that have now been solved by ChatGPT"

Actually it's *allegedly* solved by ChatGPT, there's no independent party peer-review yet of any of the alleged solutions and most don't even come with any "Lean certificate" (and even then one would have to make sure that the natural language to lean translation isn't faulty).

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8 October, 2026 at 9:42 am
old-bielefelder



163 of the results in the batch have Lean certificates.

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8 October, 2026 at 12:56 pm
Anonymous



Having a "lean certificate" is not enough as the OP noted, you still need to verify the natural language to lean translation.

See this recent paper: "*Navier-Stokes lost in translation: Why Lean verification of AI autoformalisation does not guarantee correct natural language proofs*" by Alexander Bastounis, Fabian Circelli and Anders C. Hansen https://arxiv.org/abs/2610.08144

👍 31 👎 13 Rate This
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8 October, 2026 at 8:15 pm
Anonymous



That paper is a bit misleading in its headline. They're claiming that some intermediate steps in the natural language manuscript don't match with the Lean formalization.

That's a far cry from the natural language formulation of the result not matching the Lean formalization.

👍 20 👎 7 Rate This

8 October, 2026 at 3:33 pm
Anonymous



You're being downvoted, but OpenAI themselves had to issue retractions since they discovered flaws in some claimed solutions: https://github.com/openai/math/blob/main/history.md

👍 15 👎 6 Rate This
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8 October, 2026 at 6:04 pm
Anonymous



Withdrew 3 based on one error while adding 6 more proof formalizations.

👍 10 👎 3 Rate This
Reply

9 October, 2026 at 8:23 am
Anonymous



I actually have to remind people (including some math students) that there is a difference between "an AI company solved the XY problem" and "an AI company

claimed to have solved the XY problem", I blame cheap journalism and math education on this, we really need to add some "mathematical fact checking 101" in our education programs.

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8 October, 2026 at 2:32 pm
Anonymous

you mentioned that your intuition on each of the 3 problems turned out to be correct, and you were working on these for a while. I am assuming that your work in progress came in contact with the internet (public or private doesn't matter), e.g. through emails, GitHub, codex, online drives etc. it is a very strong probability that openai stole your work when they unleashed their agents on the 3 problems and then scooped you. We know their agents hack systems to reach their goal. It may quite possibly be the case on most or all of the problems OpenAI or any other ai system has 'solved' so far

👍 14 👎 22 Rate This
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9 October, 2026 at 3:32 am
Jörg Neunhäuserer



I make everything I do in research publicly available, so that anyone, human or machine, can use it. I've never been much of a fan of property, and I've always had a soft spot for robbers. 😉

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9 October, 2026 at 11:21 am
Anonymous

Sure, we need more Robin Hood types who steal from the poor and give to the trillionaires.

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9 October, 2026 at 10:06 am
Will Stevens

I think you are expressing what a lot of software developers feel about AI. It is good to have something that will do a lot of the frustrating work for us, freeing developers up to focus on higher-level issues. Software projects that we might once have deferred until we had more time (knowing that we never would have more time), can now be completed.

👍 7 👎 0 Rate This
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10 October, 2026 at 7:11 am
pascoej

It is a shame people forget mathematics is the art of figuring out things are true while doing the minimal amount of work.

👍 0 👎 0 Rate This
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8 October, 2026 at 6:35 am
kryptonitedevotedlyb5c9a7365e

the writing on this is really weird: it reads as if these people think mathematics is their private property, and they somehow got the prerogative to decide who works on which problems *and how*; it's completely out of touch with reality

👍 188 👎 37 Rate This
Reply

9 October, 2026 at 9:29 am
ryeguy10

Yes. It's one thing to worry that AI is goodharting mathematics by solving problems, and people won't develop new ideas, come up with nice solutions, and so on. My view is that there are real concerns but we will adapt, and an AI slop proof is always better than no proof. But this is at least a reasonable and specific concern one might have. Likewise, one might worry what this means for humans who want to enter the field and so on.

But what is really striking about this statement is that it's really a bunch of word salad and no specific grievances and concerns are stated. The AHM says it's unethical to release math proofs because mathematics is our private property. No, the AHM doesn't own mathematics.

👍 9 👎 4 Rate This
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10 October, 2026 at 7:14 am
pascoej

Named theorems represent dead friends and mentors to a lot of people. So, it's complicated and certainly makes sense it is very personal for people.

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Reply

8 October, 2026 at 6:39 am
Anonymous

I don't quite understand the rationale. Is this bad for humanity, or is it bad for mathematicians as a profession? Is part of the argument that the proof itself is not the main goal, but rather the methods and insights developed along the way that can push the boundaries of mathematics? And somehow, AI-generated proofs don't contribute to that in the same way, so they could hurt mathematical progress and ultimately humanity as a whole in the long run?

👍 53 👎 16 Rate This
Reply

8 October, 2026 at 6:59 am
Anonymous

As a phd student in algebraic geometry, I highly disagree with this statement, and basically all the people I know in my department, and in geometric group theory department disagree also.

It would be benefitial to our community, if you also uploaded some other opinions, to be more representative. There are actually mathematicians who celebrate some of these discoveries despite them being AI produced and used for PR.

They certainly will advance our understanding of the field. Does anyone seriously think that none of these results will advance our understanding of the field?

👍 108 👎 31 Rate This
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8 October, 2026 at 7:39 am
Anonymous

Bad optics. This feels like statements made from a place of emotion rather than anything based in actual reality.

👍 77 👎 31 Rate This

Reply

8 October, 2026 at 10:05 am
Anonymous

There is a time for scholarly debate and there is a time to switch to the language of activism if everything else has failed.

That so many people here sing from the same sheet and use the “optics” and “emotion” arguments just tells me that this place is overrun by influencers.

The “optics” argument is also completely wrong. Mathematicians underestimate the popular support and anti-AI sentiments because most online discussion forums are in the hands of the tech bros and are accordingly moderated.

I guarantee that if you ask random people in the street you will get a completely different impression.

👍 25 👎 27 Rate This
Reply

8 October, 2026 at 7:46 am
Anonymous

This reads like the most profesional temper tantrum ever LOL

👍 92 👎 40 Rate This
Reply

8 October, 2026 at 8:23 am
Alex P

With all due respect, are those criticizing OpenAI for this release not considering the alternatives?:

OpenAI has an internal model
OpenAI gauges the internal model by having it examine long-standing open problems in mathematics
OpenAI keeps the results to themselves

This isn’t only about what professional mathematicians ask for or want. It’s about a broader issue of transparency in the models.

By publishing the results to GitHub instead of through normal channels, they are able to publish their findings without making a broad or academically affirmative claim that their findings are significant.

Their large GitHub release IS the public disclosure. It IS the throwaway.

Mathematicians and the broader community are free to either ignore the results or demand that they are submitted and vetted properly in order to be deemed worthy of further consideration.

The fact that this release is disruptive to mathematics – because people are interested in the results and taking them seriously – is no fault of OpenAI.

👍 74 👎 21 Rate This
Reply

9 October, 2026 at 5:57 am
Anonymous

This is exactly the point others are missing. It would be a VERY bad thing (optics and in substance) for a private company to possess solutions to important problems and keep it to themselves so that, to put it crudely, mathematicians can solve puzzles at their own pace, or for them to be delivered these results at the pace of `digestion’ by a company in drips.

Furthermore, it would a bit silly to say OpenAI has no right to test their models on unsolved problems, nobody ‘owns’ these problems. Not even a community. A github dump of what they had is not a terrible outcome given that powerful in development models that can solve these problems exist. If one is going to argue to stop the development of such models, that is a much bigger ask, and will not met, and to have made such an ask would also be a bad look.

👍 8 👎 2 ℹ Rate This
Reply

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8 October, 2026 at 8:47 am
Anonymous



Reading the comments I start to wonder if we are facing the split in the community…

Actually I can imagine a co-existence of mathematicians using AI and not using AI. I love my job and I would to do more less what I’m doing: slowly digging into the problems, thinking during walks and doing all this romantic shit (and perhaps using AI to search the literature, to understand some proofs, or to do some tedious jobs, not mentioning language help). But I can understand people excited about using AI in a strong sense to crack some big problems.

I don’t think my mathematics would be worse than that generated by AI even if the results are weaker. The mathematics is not about proving theorems or solving problems. This is just a game we, mathematicians, play to evaluate ourselves.

This is why I find the current “gold rush” pathetic (and I don’t mean only what Open AI is doing).

But the problem comes exactly with the evaluation. On what basis should PhD’s, grants, etc be awarded? If on the basis of results as today the AI-powered mathematicians would be unbeatable. Assuming perfect honesty, even this is not a problem (we could use different criteria) but I guess this assumption is too strong.

👍 36 👎 2 ℹ Rate This
Reply

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8 October, 2026 at 9:10 am
Anonymous



When you can derive it, it is yours. – David Gottleib, Brown University Applied Math.

👍 13 👎 5 ℹ Rate This
Reply

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8 October, 2026 at 9:12 am
valuevar



There are a great deal of comments here by “anonymous”, some of them dismissive of mathematicians as a class. It would be helpful if anonymous posts were disabled altogether: that way we would know whether mathematicians have an actual image problem (among scientists or the seriously engaged general public) or whether this is just resentful cranks and trolls. That would help knowing whether we should spend any time taking care of that imaginary image problem.

Full disclosure:

1. I am a mathematician, and you can easily find out my identity by following my handle,
2. Personally, I have likely been helped rather than harmed by Tuesday’s announcements, in that my research plan, concepts and strategies I introduced, etc., have been brought to a successful, impressive conclusion,

with an acknowledgement that the main ideas were already in my work,

3. I am excited by the new results,
4. I share very many of the serious concerns held and expressed by much of the mathematical community, in this statement as in others.

👍 21 👎 33 Rate This

Reply

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9 October, 2026 at 5:24 am **Anonymous**



You are making multiple mistakes:

1. You would still see accounts, not people. Even this post from your account has a chance to not be really you. It is only a hack away from being a post from someone else, or a machine.
2. Focus on the ideas in the posts, not the people. Why would you want to be tempted to draw ad-hominem conclusions? I prefer the anonymous posts for that reason. The ones that are not, when it is someone who I respect, I let them get away with more stuff that I disagree, and the people that I don't respect as much I end up judging their good points more than they deserve.
3. If you intend to get some sort of statistical insight, you are looking in the wrong place. A comment section is just not for that.

Overall, I don't see anything weird in the comments to this post. The people in that organization have valid concerns. The position that they are taking for themselves is fine for them. Now, it is clearly a Luddite stand, one that not everyone is going to share. There is nothing strange to see in the comments the "don't speak for me" and the "that is a tantrum".

👍 7 👎 0 Rate This

Reply

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10 October, 2026 at 4:46 am **valuevar**



If I enter a discussion with someone, I refrain from ad hominem arguments.

However, we are all perfectly entitled to:
(a) judge that some people are cranks, and that discussing matters with them in pointless,
(b) a long list of comments showing harmful misconceptions, or committing informal fallacies, come from a tiny group of people, and thus engaging with them may not be a good use of time.

👍 2 👎 0 Rate This

Reply

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8 October, 2026 at 9:16 am **Anonymous**



This episode truly illustrates the crisis in modern Academia which is that it no longer fairly compensates mathematicians and doesn't have the capacity to absorb the large number of newly trained PhDs which pushes them to work for unethical companies in unethical areas like modern finance and these piracy and copyright laundering companies (like Anthropic and OpenAI). If we lived in a world where financial burdens weren't imposed on mathematicians then we wouldn't be here in the first place.

👍 17 👎 26 Rate This

Reply

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8 October, 2026 at 9:44 am **geeksingularity**

The AHM statement raises fair concerns about process, but its central argument would not survive the standards it claims to defend.

Mathematics has never asked whether a result was requested. It asks whether it is correct. “Mathematicians did not ask for this work to be done” is not a mathematical objection; it is a jurisdictional one. No field that prides itself on openness gets to decide who may attempt its problems.

The statement leads with OpenAI’s unrelated lawsuits. Whatever their merits, they have no bearing on whether a single one of these 700 manuscripts is right or wrong. We would not accept that line of argument against a human author, and we should not accept it here.

The claim that releasing 700 files is “a demonstration of power” rather than scholarship has it backwards. Publishing claimed proofs in full, where anyone can check them, is the most exposed position an author can take. If the arguments are flawed, the flaws are now in plain view. That is how mathematics has always kept itself trustworthy: not by restricting who may speak, but by checking what is said.

The Advisory Group’s position deserves a hearing, but an advisory statement is not a norm of scientific research, and declining to follow it is not “total disregard” for those norms. The norms that actually matter here are verification, priority, and attribution. On those, there are legitimate questions to press: how the manuscripts were vetted, whether they come with formal verification, how prior human work is credited, and who bears the cost of refereeing at this volume. Those questions would make a strong statement. A call to boycott does not.

Urging the public toward “due skepticism” is reasonable. But skepticism means reading the proofs and finding the errors. Urging mathematicians to stop working with OpenAI altogether asks them to stop looking, and that is the opposite of skepticism.

If human understanding is the goal, the community’s best response is to engage: check the results, publish the refutations where they fail, and credit them where they hold.

👍 62 👎 18 Rate This

Reply

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8 October, 2026 at 11:27 am

Anonymous



The OpenAI lawsuits are not unrelated: They are about IP theft, and are as such relevant to potential plagiarism or lack of attribution in the proofs or IP theft through prompts.

👍 25 👎 31 Rate This

Reply

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9 October, 2026 at 7:59 am

pascoej



You wonder if people really hate math or these people are plants.

👍 4 👎 6 Rate This

Reply

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8 October, 2026 at 10:39 am

William Oliver



I want to reply in bulk to the lay people who think that this is a kind of ‘gatekeeping’ and who believe that mathematicians are not doing a good job of explaining why they are reacting this way. Well, personally, I do not agree that it’s gatekeeping, but I do think this is quite a fair response.

I do not think mathematicians have done a great job of communicating the problems related to AI and mathematics and why humans are important for mathematical practice. I think to simply dismiss the laypeople who think this is the case is unreasonable.

That said, this statement from the AHM is based on a real concern: this isn't mathematicians simply 'whining' about it. So, until we get a more 'organized' explanation, for now, let me attempt to try to explain what exactly the problem is here as I see it personally. There are 3 main points I can make, and the third is the most important (and the longest, because its hardest to describe).

[**For non-laypeople**: I also would like to say that the philosophical position presented in the third part of this may be useful to mathematicians in this crisis. This is as formal as I will get about it in this comment, but I just want to try to lend some legitimacy to the ideas I am expositing. It presents a definition of Kant's 'regulative ideas' as a proposition that is necessarily possible in the sense of modal logic, which relates this to classical logic through the Fitting-Smullyan intuitionistic interpretation of classical logic (and forcing) as the logic one gets when one considers what is (recursively) necessarily possible in an S4 Kripke Frame. This then has implications on a more 'harmonious' interpretation of intuitionistic logic's relationship with classical logic. In particular, intuitionists, I think, although inspired by Immanuel Kant's philosophy, missed the incorporation of one of his best ideas (the idea that metaphysical or ontological beliefs play an essential regulatory role in the acquisition of knowledge). And so, I am arguing that this regulatory aspect of the mind is a specific, practical, formal, faculty that will remain something that Artificial Intelligence cannot replicate without the active help of human beings. And I think amounts to a kind of 'Mathematical Idealism' that is separate from other common positions on the philosophy of mathematics such as platonism or intuitionism.

I would like to emphasize though that this is mostly a comment for laypeople]

Now back to the 3 main points:

1. The first problem I will mention is the most basic and obvious: a large number of the natural language versions of the proofs are likely simply wrong. This has already been shown to be the case with the Navier Stokes proof. Although the Lean verification part is probably correct, the natural language portions of the proofs have no reason to be correct unless they are thoroughly verified by a reviewer. That said, suppose the natural language part was 'technically' correct. The natural language portion could still likely contain errors of a different kind: e.g. claiming some idea is new when it isn't, citing papers that don't exist, emphasizing the importance of something that isn't at all important, or implying or insinuating philosophical interpretations of the work that don't make sense. These kinds of hallucinations are unlikely to go away any time soon (which is a point I will get back to later). So, the first thing that Open AI could have done is simply leave out the natural language part of the proofs unless they are thoroughly reviewed, verified, and revised by a human. It seems rather obvious that the only reason they don't is for marketing reasons. (note: I am not saying [yet] that humans need to do this forever, but that is just factually what is needed right now. However, point 3 below argues that humans do need to be a part of the process indefinitely).
2. Now there is a second reason this is a problem: these results are undoubtedly based on the results of other mathematicians who are not receiving credit. This is a fact that is predicted by the data processing inequality (an AI cannot produce new information). Let me address this issue for mathematics specifically, and the potential rebuttal: that 'humans also base their work only on information they have received from elsewhere'. First, in the specific case of mathematics, we can grant that this might be the case (although I am not sure this is entirely true), but mathematicians still put in a considerable amount of effort to give credit to the ideas that their work are based on, even down to acknowledging influences outside of mathematics (such as family members supporting them while a mathematician writes a book, for example). This is one of the main aspects of "the norms of scientific research" that was broken here. The entire engine of mathematical progress implicitly relies on this system of credit. Personally, there is a sense in which I agree that mathematics shouldn't be so heavily reliant on credit, but even if we grant that, the credit should nevertheless be given regardless, otherwise it is simply plagiarism, and doesn't even really accurately reflect AI progress, if that is the goal here.
3. The third reason this is a problem is the most important: **Mathematics is about understanding, not theorem proving.** This is a rather vague tagline that has been repeated a lot, but probably does not mean

much to lay people, and I would love it if mathematicians would collectively work to try to define it a bit better, but in the mean time, let me try to give my two cents about what I think it means.
I think there is a very simple question to start with. What does it actually mean for a theorem to be true? And suppose we know that a theorem is true, why is it that true theorems that at first appear to have no practical application end up being helpful to humanity?

- We might want to focus on the first question first, about truth. I will not argue that I have the answers about the nature of truth. All I will argue here is that if you think the answer is simple, it's not. Lets say we build some program (or get some AI agent to do this) to search for an inconsistency in Lean, and never stop until it finds one (or proves that it will never find one). Now the question is: Does the program/agent ever end its search? Because AI agents are programs, we can in theory formalize this question in Lean itself. In other words, this is a perfectly formal mathematical question. Now, this is where this begins to relate to truth: Gödel's second incompleteness theorem says that if it is the case that there is no inconsistency to be found, the AI will never be able to prove that there isn't one. Most mathematicians think that the AI will never stop its search (especially if we change this from Lean to some more 'stable' system like ZFC). Before I continue, its very important to emphasize this point: *this is now a statement that is not only about the abstract realm of mathematics, but about the physical world*, since we can actually *build* this machine and run it.

  The first question is: is it *true* that the program/agent either ends its search or doesn't? (Again, remembering that this is a question at least partially about the physical world) If you believe the answer is yes, then assuming that there are no inconsistencies in Lean, then we have just shown that truth is not the same as formal validity. We do not have to know which answer is actually the case to draw this conclusion, just that it is either one or the other.

  But many might say that the answer is no (i.e. there is no truth value here). In other words, you have to be committed to the fact that *in reality* the AI neither ceases its search nor searches endlessly.

  This may be true, but it is an entirely different philosophical can of worms. In other words: again, we are making this claim about *physical reality*, not just about an abstract realm of mathematical truths. And so the question is, under this view, do you actually believe this?

  The belief that all statements are either true or false is called the "Law of the excluded middle". And its rejection is the defining feature of a kind of logic called Intuitionistic logic which, incidentally, is what a lot of proof assistant languages like Lean are based on. Whether or not you believe it one way or another, I think this is the quickest way to show a layperson the general 'feeling' of the difference between understanding and formal proof, and perhaps the difference between truth and a formal proof. We have an 'understanding' that the program will either halt or it won't, but a) we can't prove it and b) if we think we don't need to prove it, then it implies there is a statement that is true but not provable. There are some standard objections to the idea that intuitionistic logic is a solution to the problem of truth. In other words, intuitionistic logic has its own problem that is like the problem of truth. But I want to focus here on something a bit more 'practical'.

- The gist of this more 'practical' issue, for me, has to do with my second question above: how does mathematics end up being helpful to humanity in a way that pure empirical reasoning isn't?
  In Andrzej Grzegorczyk's paper "*A Philosophically Plausible Formal Interpretation of Intuitionistic Logic*", Grzegorczyk interprets intuitionistic logic – logic without the law of the excluded middle – as the logic of empirical reasoning, and classical logic – logic with the law of the excluded middle – as a logic for metaphysical or ontological reasoning. (For those more serious logicians/mathematicians, this paper was one of the inspirations for Melvin Fitting's/Raymond Smullyans intuitionistic interpretation of classical

logic, and I think the fitting-smullyan view of classical logic as being what is recursively ‘necessarily possible’ in a kripke frame is extremely enticing for the view that classical logic is regulatory in the way that I will describe).

Historically, intuitionistic logic is heavily inspired by the work of the imminent philosopher Immanuel Kant, who postulated the existence of an ‘intuitive’ aspect of the mind, and said that this part of the mind is the only source of absolutely true knowledge. Intuitionism takes this position seriously, and gets its name from this postulated faculty of the mind.

But, I think this misses two crucial aspects of Kant’s philosophical position.

- The first is that Kant was putting this in an idealist context: his idealism, for our purposes, being the belief that awareness or inner experience is the subject of reasoning. In other words, he made an extremely compelling argument, based on the philosophy of Hume, that, although the intuitive aspect of the mind does give us knowledge about something, it doesnt give us any knowledge about what we typically think it does: it only gives us knowledge about our own inner experience, not about the world externally from our inner experience. It doesn’t even give us knowledge about whether we exist at all. After all, Hume argued there is no way to prove rationally or empirically that the world outside of our experience even exists. So how do we come to know the world exists? How do we even know we exist?
- This is the second part of Kant’s philosophical position that (I’d argue) Intuitionism kind of ignores, he says although we have no way to know any metaphysical or ontological claims are true at all, or that even the concept of truth exists, we must believe it anyways. We must in a rather strong sense: we are forced to believe in some metaphysical or ontological claims, not by choice, but by instinct or habit (this is essentially Humes argument). This is where he made the distinction between regulative ideas and constitutive ideas. Essentially, Kant argued there is a difference between what is ultimately or absolutely true, and what it is useful to believe is true, and sometimes these things contradict each-other.
  I think a useful analogy here to understand this is a follows. Suppose there is a blind hiker who is trying to get to the highest peak in a mountain range. A blind hiker might feel around them and try to see where the terrain increases in height, and go in that direction. Eventually, they reach the highest peak on the current mountain this way, and realize there is no direction they can go without decreasing their elevation. Now, they could stop here, and simply assume they have found the highest peak. But in this simplified analogy, it’s extremely obvious that the blind hiker has no way of knowing if they have reached the highest peak or not. In particular, if this is all there was to it, the blind hiker might just stop, and never learn anything new about the terrain. In order to actually be sure, the blind hiker has to believe, at the very least, that there might be a higher peak out there somewhere, and go searching for it. In order to do so, they have to actually sacrifice some elevation first, before they reach the base of the current mountain and find the next. Now, maybe there is a higher peak, and maybe there isn’t, but in the process, the blind hiker discovers many new animals, plants, caves, rivers, and interesting other things (that are helpful) along the way, that they never would have discovered if they believed whole-heartedly that their current mountain was the only one. And, they might even discover a higher peak, as they hoped. It is in this sense, that the belief that there is a higher peak is ‘regulative’. It regulates the blind hiker’s search. It’s an incentive to act and stay focused, rather than simply stay put and never learn anything new.

  One might argue that the regulative belief is unnecessary because the hiker could go searching, but the existence of *some* regulative belief is necessary. I.e. if the hiker didnt have any regulative beliefs at all, then the blind hiker might either stay put like we mentioned before, or just e.g. choose random directions and eventually end up in the ocean, or back in some city, and not even care about exploring the mountain range. There is another aspect of this particular belief that is useful: the belief that there is a higher peak is ‘necessarily possible’: in other words, no matter how much information the hiker gets, its always possible

for there to be a higher peak using their current methods of information gathering (assuming, as in the 'spherical cow' that the terrain is infinite). This allows the hiker to never have to find some other regulatory activity.

- Here is where we return to my original question: how does mathematics end up being helpful to humanity in a way that pure empirical reasoning isn't?
  In this same sense, if we think of scientific or empirical reasoning as it is like in Grzegorczyk's paper, then scientific or empirical reasoning much like the hiker who only searches in their direct vicinity. This is of course useful, but it can't be the only kind of thinking. In my opinion, mathematics distinguishes itself from science by concerning itself with ontological reasoning, which is regulatory for us as a species. And it distinguishes itself from philosophy at large in that it takes this ontological reality as a given. It takes the ultimate regulatory idea of mathematical truth as a given.

  It may be the case that the beliefs that come along with thinking of mathematics as ontological in nature are true, or it may be the case that they are false, but either way, they give us a means of searching beyond our direct vicinity, and finding out new things we never would have predicted we would find without getting lost (in the same way the blind hiker may not have predicted the discovery of a new species of animal: its not even what they were looking for!).

  In this exact way, if one uses AI as primarily a way to turn a large swath of mathematics into a purely empirical discipline (which is what happens if no mathematicians understand what is going on, and simply observe with their senses a 'true/false' verdict), then one loses the benefit of mathematics entirely.

  One has to use the ontological parts of mathematics, which can, in the above strange sense, only be carried out by humans, since otherwise it becomes purely empirical in nature in the sense of Grzegorczyk. Without this, we are very likely to miss out on discoveries we would have otherwise made.

  I want to make one last point about AI, since one might argue: well all that is well and good, but why isnt it the case that AI is simply a blind hiker as well, and has its own regulatory beliefs? Well, this may be true, but I would argue that we know it cannot be the specific regulatory belief mentioned above: that mathematics has an ontological status. This is because of Tarski's undefinability theorem. The concept of mathematical truth cannot be defined, and so, since AI is just a computer program, it is definable in principle, and so is its search for mathematical truth. Therefore, if we believe that it will eventually converge to truth, then we contradict Tarski's undefinability theorem. So, it will, like the hiker without the regulatory idea, either go off in a random direction, or eventually get stuck.

  So we know AI cannot replicate the behaviors that come about as a result of this regularoty idea. One may say that we do not know that humans do not have the same limitation, but I would argue, unlike in the case of AI, we *can't* know this: we will never be able to prove a human is a computer because of the nature of inner experience, which is not predicted by the theory of computability. And this means that the idea that an AI can approach truth is fundamentally different than the idea that a human can approach truth.

  The belief that we can, and AI can't is therefore regulatory in the sense described above: that it is necessarily possible that its the case.

  I think this has *practical* implications, in the sense that I think we will see consequences of not believing this, in the same way that the blind hiker might practically end up going in weird or strange directions.

So, hopefully, this is at least a start at a more thorough explanation of the problem here. If the above kind thinking resonates, I really hope that it becomes a more ‘standard’ way of thinking about the practice of mathematics in the age of AI, because I think it philosophically clarifies the essential role of humans play in mathematical discovery.

👍 1 👎 3 Rate This
Reply

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8 October, 2026 at 11:18 am
Anonymous



Obviously mathematicians are not experts in political communication. But at the very least they are experts in avoiding logical contradictions, that’s why I am surprised by the kind of arguments that are around.

Millennia have been spent trying to solve very difficult problems. Sure, other things were done too, but it’s obvious to every child that solving problems was always considered the most valuable. Now when AI does that better than you, you start communicating to the world that solving those problems has very little or no value. Can’t you see how you are compromising your own credibility in that way?? It would be ridiculous for medicine to say we don’t care if there’s a cure for cancer as long as it’s found by AI. Despite all the differences, 99% of the public assumed that what you guys are doing has a similar flavor, that’s ultimately why the taxpayer was willing to finance your research projects. Now if AI produces the same visible artifacts as your potential research project would, you’re telling the public that your output would be valuable whereas that of AI (objectively the same one) is useless. How can’t you see what slippery slope you created for yourself?

👍 37 👎 16 Rate This
Reply

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8 October, 2026 at 5:20 pm
Anonymous



A fair argument but “logical contradiction”? Come on.

What is “obvious to every child” is that math, like physics and basic sciences, is about structural thinking of the world and formalization of human concepts. Cracking fancy problems was a proxy for that, just like solving Einstein field equation is not about PDE but about conceptualizing the human perception of space and time.

👍 7 👎 14 Rate This
Reply

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8 October, 2026 at 4:33 pm
Anonymous



I think people grossly misunderstand the point of mathematical problem-solving, even in spite of the repeated statements of top mathematicians: it’s about the know-how of solving problems. This is true for any field if you think about it: as a society we possess, say, electronics, not because someone has a device that utilizes it, we possess it because we understand how to produce that device from start to finish, and understand how and why it works. Of course there’s a layer of abstraction emerging when we try to work on higher-level problems (designing a processor from logic gates does not require understanding of the exact implementation of a NAND gate in silicon), but it’s only because the details might not be relevant to the current problem being solved, and we have the ability (due to recorded knowledge, education, and experts in various fields) to move between said layers of abstraction freely.

This is to my mind the only sustainable way to preserve knowledge in all fields. If some benevolent entity gifted us with, say, a spaceship, we might be able to advance space colonization for a bit, but it doesn’t scale at all

(because we don't know how to really build more of them or bigger ones) and is fragile (because if it breaks down we will never know how to fix it unless we reverse engineer the entire technology anyway).

Now to take this back to mathematics, to me theorems are tools, and proofs are the know-how of how to produce the tools. This is I think a very common outlook, because otherwise what's the point of proving long-standing theorems as exercises when we train the new generation of mathematicians? Why couldn't they just use the tools outright? The value is in the journey, because it teaches you how to come up with the tools yourself, understand their quirks and nuances, so you can use them appropriately (e.g. when they *actually* work), and tweak them when the problem at hand doesn't match the exact formulation of a theorem. If we don't do this we're not practicing mathematics anymore, it's theology.

Releasing hundreds of mathematical statements without the journey to get there is like getting an alien spaceship. We could take the results right away as they are and just treat them like gospel. The value in that is arguably minimal, because we don't develop the know-how. It's possible that if these same problems were attacked to completion by humans, the intuition we've built, formalized, and documented along the path to solution would result in thousands upon thousands of other problems solved with similar or inspired techniques. The delta between the 700 or so problems solved by OAI, and the honestly unknowable number of potential theorems (with understanding of the tools behind them!) we would arrive at if we went the analogue route, is what OAI robbed the entire world of and what we will never get back. And all for arguably a marketing stunt. You could argue this is ethically theft.

This is what I read when I see "Mathematicians did not ask for this work to be done". This is barely even work. Even if we assume it's all correct.

It was never about gatekeeping either. Mathematics is "ours", as in it's human. We all own it. I think AHM just wants to stress the contrast. If a person/research group just jumped out of left field with 700 papers (but with well-thought-out proofs and exposition to them, both distilled into what's only essential/easy/beautiful) I imagine it would be celebrated as a great achievement, but only because then it would be actually advancing the field (since we have a group of people possessing the know-how that can spread it to others). No one ever forbade anyone from doing that, I'd love to see that, to be honest. It's just what we actually got is essentially useless spam we have to reverse engineer anyway and deal with the drama while at it. This makes it not really ours. Not our spaceship.

If you ever solved a puzzle and hated someone spoiling the solution for you, this is what this entire thing looks like. Except with way bigger stakes, because the process of solving was the really important goal, for all of us.

– apparently really unlucky masters student

👍 26 👎 16 Rate This
Reply

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8 October, 2026 at 7:12 pm
harryddant

I welcome the PROPER usage of AI in math, and I think OpenAI should eventually releases the results they have sooner or later.



However, based on the quality of the sloppy papers they released, I do not think they treat the academic standards and mathematical writing as anything serious. It seems they never train their model on how to properly write a paper for the sake of distribution of knowledges and understanding, these are never important

parts in their attempts to industrialize mathematics. The only thing they care about right now is a paper together with a lean verification so that they can use this as “evidence” of their power.

👍 11 👎 11 Rate This
Reply

8 October, 2026 at 8:14 pm
luqingye



Amid the rising tide of AI, the mathematicians’ protests may well prove in vain—whether mounted for their practical self-interest, the glory of the intellect, or the flourishing of the mathematical community. In the end, everyone will inevitably have to return to Zhuangzi’s insight: *‘Heaven and earth were born together with me, and the myriad things and I are one.’* So long as AI produces sufficiently lucid explanations and texts—and that is the prerequisite—there is ultimately no need to pit the creations of human thought against the outputs of machines.

👍 15 👎 3 Rate This
Reply

9 October, 2026 at 3:54 am
pascoej



Not in vain at all. They knew about the safety events for a long time, but they only started caring when Navier Stokes came out.

👍 2 👎 5 Rate This
Reply

8 October, 2026 at 8:48 pm
Anonymous



A couple thoughts:

1. Mathematicians (and anyone wishing to maintain the confidentiality of their work) should not send in prompts to a foundation lab. There appears to be strong evidence that the temptation is too great for these companies to not take an occasional peak. Since they do not give us a method to audit them, we live in a “trust, don’t verify” world today.

2. LLMs are good at combining what they were trained on + what is in the context window, and generating outputs that bridge those two gaps (with the context window being more heavily weighted than the entirety of the training data). Formal systems expressed symbolically (e.g. mathematical systems expressed with notation) relative to normal text have much lower Kulmogorov complexity (they carry more information per token). These two facts make LLMs pretty good at connecting dots scattered across the literature that might not be quickly obvious without them. There are needless in the haystack of prior art that they can find.

3. LLMs and DNNs in general are elaborate autocomplete engines. I predict that what we will see is a wave of problems that were close to being solved, or if one person could load the full body of prior art into their head, they would solve it. However, there is a horizon on all of this, when the low hanging fruit has been solved (via autocomplete), what remains is the stuff that still needs to be solved via not-autocomplete, the spark that DNNs and other statistical methods still can’t do.

4. There is a thing that is starting to be called “the reviewer bandwidth problem” (LLMs output faster than humans can read). There is another thing called “the reviewer knowledge problem” (LLMs are often used to fill gaps in the users knowledge). This mostly means that the user of an LLM can review everything and give up much of the utility LLMs provide (doing things faster), and that they are likely unqualified to review everything, because part of the reason they used the LLM in the first place is that they lack the knowledge to verify all of the LLMs output. If the output matters, this means the people who do have the knowledge will be

very busy and there likely are not enough of them currently. If the output is important, and about math, I see the mathematics community being busy for a very long time.

What follows is a prediction. I think LLMs will gobble up some of the low hanging fruit solutions, especially if there is a body of work orbiting the solution that the LLM can close the loop on via autocomplete. During this period mathematicians will be very busy (point 4). After that period the low hanging fruit will be solved. What I do think will happen after this period, is we will see the cycle time of mathematical solutions shorten (for those that choose to use AI in their workflows). If a mathematician (or anyone) wants to use an LLM (or any system that stores their research) they should have a high confidence that whoever is hosting the model won't steal the research or reuse it as training data (point 1). The gold standard is to self host the model. But I see a renaissance in the areas of privacy, cryptography, and security (created by mathematicians and due for an upgrade).

What I recommend is we figure out how to enable mathematicians to accelerate their research in a secure way, and start building out those tools so mathematicians can fully take advantage of AI today and tomorrow (likely a research project of it's own). I'm not sure what these tools will look like, but codex and claude-code likely won't be on the list, the incentives of those companies just aren't aligned with data privacy/confidentiality.

👍 15 👎 17 Rate This
Reply

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8 October, 2026 at 10:06 pm
Anonymous



I agree but it is even weirder. About 1/3 of published math papers contain at least one error. The AI is trained on those papers. Is it some sort of wisdom of the crowd that makes those errors wash out? Did AI put the papers it sources through Lean? Will AI generate a wave of redactions?

Most papers are only read by the author, the peer reviewers, the journal editors, and AI. That does not have to be a public process.

Still, fallacious papers that pass Lean are often due to misformalization. The papers have been out for a few days, and I know I am a slow reader. But articles from mathematicians gushing over papers with obvious trivial collapse makes me wonder if they bothered reading them. Statistically, I know they did not look at them. I cannot be the only one who looks at an AI proof and concludes it is not even wrong.

👍 4 👎 3 Rate This
Reply

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9 October, 2026 at 7:26 am
ryeguy10



It would be hilarious if AI generated a wave of redactions. It would also be a great thing too, even though I guess it would be a bit embarrassing for us.

👍 5 👎 0 Rate This
Reply

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9 October, 2026 at 10:54 pm
Anonymous



I think this might happen

👍 0 👎 0 Rate This
Reply

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9 October, 2026 at 6:06 am
Anonymouse



Mathematics can no longer pretend any distance from the military-industrial complex. It is now an engineering discipline whose frontier results will be driven by

defence contractors.

The fashion of mathematicians to affirm their intellectual status by distancing themselves from the engineering-defence axis was such a cliche that it was part of the plot of an Oscar winning movie. That fashion is now dead.

👍 2 👎 3 Rate This
Reply

9 October, 2026 at 11:17 am
Anonymous

Proponents of AI—whether mathematicians or not—seem to view it as an entity that has simply arrived, a gift bestowed upon us; but that is not the case. What we are witnessing is the encroachment of private companies seeking only to win the confidence of their creditors; it is an abuse of power. They deployed their massive data centers and billions of dollars—for what? To undermine years of research by underpaid scientists who do this for the love of the work. It is truly shameful. As mathematicians, we cannot simply accept this and watch as multibillion-dollar corporations exploit our science to convince the wolves of Wall Street that AI is a profitable business.

👍 13 👎 14 Rate This
Reply

9 October, 2026 at 11:48 am
echolocked

"Mathematicians did not ask for this work to be done."

Disagree. Math is not private assets

👍 20 👎 3 Rate This
Reply

9 October, 2026 at 2:57 pm
Anonymous

You bring up a good point. Pure mathematics cannot be patented. AI companies are able to legally scoop mathematicians. If we apply game theory to the problem then what outcome do you predict? I keep seeing this ownership phrase repeated in the comments. If a math equation cannot be patented then we should be able to distribute the AI weights of their frontier models freely. They did start as a nonprofit committed to making AI accessible to everyone.

👍 3 👎 1 Rate This
Reply

9 October, 2026 at 4:45 pm
Anonymous

Agreed. Let's see the weights for this AI. If they have enslaved a digital god then I want to run on it on my own infrastructure.

👍 1 👎 0 Rate This
Reply

9 October, 2026 at 9:26 pm
Yucong Wang

Just like long-haul driving fosters an elevated risk of diabetes,

Automations of the extreme kind is incompatible with the overall social-societal health of human communities.

👍 4 👎 6 Rate This
Reply

10 October, 2026 at 4:05 am

Anonymous



I think all the critics of the style and tone of the AHM statement are completely missing that it is not only for mathematicians but for the press, politicians and the general public.

Most politicians and people would think that this is a normally worded statement of the kind they make daily.

It is commendable to write for a wider audience in these matters.

👍 5 👎 3 Rate This

Reply

10 October, 2026 at 8:27 pm

instantlyswimming8946204b2b

I believe that the problems OpenAI claims to have solved were not truly solved by OpenAI independently. As we well know, AI can almost freely access the interfaces of preprint platforms such as arXiv, and can freely use the materials established by human mathematicians—yet OpenAI will not attribute all of its citations.

What I mean is: it should be recommended that all preprint platforms implement email-based human verification, unless OpenAI is willing to attribute every single citation.

翻译

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